

12

Pipeline Economics

In this chapter we discuss the cost of a pipeline and the various components that contribute to the economics of pipelines. These include the major components of the initial capital costs and the recurring annual costs. We also examine how the transportation charge is established based on throughput rates, project life, interest rate, and financing scenarios.

12.1 Economic Analysis

In any pipeline investment project we must perform an economic analysis of the pipeline system to ensure that we have the right equipment and materials at the right cost to perform the necessary service and provide a profitable income for the venture. The previous chapters helped determine the pipe size, pipe material, pumping equipment, etc., necessary to transport a given volume of a product. In this chapter we analyze the cost implications and how to decide on the economic pipe size and pumping equipment required to provide the optimum rate of return on investment.

The major capital components of a pipeline system consist of the pipe, pump stations, storage tanks, valves, fittings, and meter stations. Once this capital is expended, and the pipeline has been installed and the pump

station and other facilities built, annual operating and maintenance costs for these facilities will be incurred. Annual costs will also include general and administrative (G&A) costs including payroll costs, rental and lease costs, and other recurring costs necessary for the safe and efficient operation of the pipeline system. The revenue for this operation will be in the form of pipeline tariffs collected from companies that ship products through this pipeline. The capital necessary for building this pipeline system may be partly owner equity and partly borrowed money. There will be investment hurdles and rate of return (ROR) requirements imposed by equity owners and financial institutions that lend the capital for the project. Regulatory requirements will also dictate the maximum revenue that may be collected and the ROR that may be realized, as transportation services. An economic analysis must be performed for the project taking into account all these factors and a reasonable project life of 20 to 25 years, or more in some cases.

These concepts will be illustrated by examples in subsequent sections of this chapter. Before we discuss the details of each cost component, it will be instructive and beneficial to calculate the transportation tariff and cost of service using a simple example.

Example Problem 12.1

Consider a new pipeline that is being built for transporting crude oil from a tank farm to a refinery. For the first phase (10 years) it is estimated that shipping volumes will be 100,000 bbl/day. Calculations indicate that a 16 in. pipeline 100 miles long with two pump stations will be required. The capital cost for all facilities is estimated to be \$72 million. The annual operating cost including electrical power, O&M, G&A, etc., is estimated to be \$5 million. The project is financed at a debt/equity ratio of 80/20. The interest rate on debt is 8% and the rate of return allowed by regulators is 12%. Consider a project life of 20 years and overall tax rate of 40%.

- (a) What is the annual cost of service for this pipeline?
- (b) Based on the fixed throughput rate of 100,000 bbl/day, and a load factor of 95%, what tariff rate can be charged within the regulatory guidelines?
- (c) During the second phase, volumes are expected to increase by 20% (years 11 through 20). Estimate the revised tariff rate for the second phase assuming no capital cost changes to pump stations and other facilities. Use an increased annual operating cost of \$7 million and the same load factor as before.

Solution

(a) The total capital cost of all facilities is \$72 million. We start by calculating the debt capital and equity capital based on the given 80/20 debt/ equity ratio.

$$\text{Debt capital} = 0.80 \times 72 = \$57.6 \text{ million}$$

$$\text{Equity capital} = 72 - 57.6 = \$14.4 \text{ million}$$

The debt capital of \$57.6 million is borrowed from a bank or financial institution at the 8% annual interest rate. To retire this debt over the project life of 20 years, we must account for this interest payment in our annual costs.

In a similar manner, the equity investment of 14.4 million may earn 12% ROR. Since the tax rate is 40%, the annual cost component of the equity will be adjusted to compensate for the tax rate.

$$\text{Interest cost per year} = 57.6 \times 0.08 = \$4.61 \text{ million}$$

$$\text{Equity cost per year} = 14.4 \times 0.12 / (1 - 0.4) = \$2.88 \text{ million}$$

Assuming a straight-line depreciation for 20 years, our yearly depreciation cost for \$72 million capital is calculated as follows:

$$\text{Depreciation per year} = 72 / 20 = \$3.6 \text{ million}$$

By adding annual interest expense, annual equity cost and annual depreciation cost to the annual O&M cost, we get the total annual service cost to operate the pipeline:

$$\text{Total cost of service} = 4.61 + 2.88 + 3.60 + 5.00 = \$16.09 \text{ million/yr}$$

(b) Based on the total cost of service of the pipeline, we can now calculate the tariff rate to be charged, spread over a flow rate of 100,000 bbl/ day and a load factor of 95%:

$$\text{Tariff} = (16.09 \times 10^6) / (365 \times 100,000 \times 0.95) \text{ or } \$0.4640/\text{bbl}$$

(c) During phase 2, the new flow rate is 120,000 bbl/day and the total cost of service becomes:

$$\text{Total cost of service} = 4.61 + 2.88 + 3.60 + 7.00 = \$18.09 \text{ million/yr}$$

And the revised tariff becomes:

$$\text{Tariff} = (18.09 \times 10^6) / (365 \times 120,000 \times 0.95) \text{ or } \$0.4348/\text{bbl}$$

12.2 Capital Costs

The capital cost of a pipeline project consists of the following major components:

- Pipeline
- Pump stations
- Tanks and manifold piping
- Valves, fittings, etc.
- Meter stations
- SCADA and telecommunication
- Engineering and construction management
- Environmental and permitting
- Right-of-way acquisition cost
- Other project costs such as allowance for funds used during construction (AFUDC) and contingency

12.2.1 Pipeline Costs

The capital cost of a pipeline consists of material and labor for installation. To estimate the material cost we will use the following method:

$$\text{Pipe material cost} = 10.68(D-t) \times 2.64 \times L \times \text{cost per ton}$$

or

$$\text{PMC} = 28.1952 L(D-t)t(\text{Cpt}) \quad (12.1)$$

where

- PMC=Pipe material cost, \$
- L=Pipe length, miles
- D=Pipe outside diameter, in.
- t=Pipe wall thickness, in.
- Cpt=Pipe cost, \$/ton

In SI units, Equation (12.1) can be written as

$$\text{PMC} = 0.02463 L (D-t)t (\text{Cpt}) \quad (12.2)$$

where

- PMC=Pipe material cost, \$
- L=Pipe length, km
- D=Pipe outside diameter, mm
- t=Pipe wall thickness, mm
- Cpt=Pipe cost, \$/metric ton

Since the pipe will be coated, wrapped, and delivered to the site, we will have to increase the material cost by some factor to account for these items or add the actual cost of these items to the pipe material cost.

Pipe installation cost or labor cost is generally stated in \$/ft or \$/mile of pipe. It may also be stated based on an inch-diameter-mile of pipe. Construction contractors will estimate the labor cost of installing a given pipeline based on a detailed analysis of the terrain, construction conditions, difficulty of access, and other factors. Historical data is available for estimating labor costs of various-size pipelines. In this section we will use approximate methods; these should be verified with contractors taking into account current labor rates, geographic and terrain issues. A good approach is to express the labor cost in terms of \$/inch diameter per mile of pipe. Thus we can say that a particular 16 in. pipeline can be installed at a cost of \$15,000 per inch-diameter-mile. Therefore, for a 100 mile, 16 in. pipeline we can estimate the labor cost to be

$$\text{Pipe labor cost} = \$15,000 \times 16 \times 100 = \$24 \text{ million}$$

Based on \$/ft cost this works out to be

$$\frac{24 \times 10^6}{100 \times 5280} = \$45.50 \text{ per ft}$$

[Table A.14](#) in Appendix A shows a summary of typical labor costs. It must be noted that these numbers are approximate and must be verified with the contractor for a specific geographic location and level of construction difficulty.

In addition to labor costs for installing straight pipe, there may be other construction costs such as road crossings, railroad crossings, river crossings, etc. These are generally estimated as a lump sum for each item and added to the total pipe installation costs. For example there may be ten highway and road crossings totaling \$2 million and one major river crossing that may cost \$500,000. For simplicity, however, we will ignore these items for the present.

12.2.2 Pump Station

To estimate the pump station cost a detailed analysis would consist of preparing a material take-off from the pump station drawings and getting vendor quotes on major equipment such as pumps, drivers, switchgear, valves, instrumentation, etc., and estimating the station labor costs.

An approximate cost for pump stations can be estimated using a value for cost in dollars per installed horsepower. This is an all-inclusive number considering all facilities associated with the pump station. For example, we

can use an installed cost of \$1500 per HP and estimate that a pump station with 5000 HP will cost

$$\$1500 \times 5000 = \$7.5 \text{ million}$$

In the above we used an all-inclusive number of \$1500 per installed HP. This figure takes into account all material and equipment cost and construction labor. Such values of installed cost per HP can be obtained from historical data on recently constructed pump stations. Larger HP pump stations will have smaller \$/HP costs while smaller pump stations with less HP will have a higher \$/HP cost, reflecting economies of scale.

12.2.3 Tanks and Manifold Piping

Tanks and manifold piping can be estimated fairly accurately by detailed material take-offs from construction drawings and from vendor quotes.

Generally, tank vendors quote installed tank costs in \$/bbl. Thus if we have a 50,000 bbl tank, it can be estimated at

$$50,000 \times \$10/\text{bbl} = \$500,000$$

based on an installed cost of \$10/bbl.

We would of course increase the total tankage cost by a factor of 10–20% to account for other ancillary piping and equipment.

As with installed HP costs, the unit cost for tanks decreases with tank size. For example, a 300,000 bbl tank may be based on \$6/bbl or \$8/bbl compared with the \$10/bbl cost for the smaller, 50,000 bbl tank.

12.2.4 Valves and Fittings

Valves and fittings may also be estimated as a percentage of the total pipe cost. However, if there are several mainline block valve locations that can be estimated as a lump sum cost, we can estimate the total cost of valves and fittings as follows: A typical 16 in. mainline block valve installation may cost \$100,000 per site including material and labor costs. If there are 10 such installations spaced 10 miles apart on a pipeline, we would estimate cost of valves and fittings to be \$1.0 million.

12.2.5 Meter Stations

Meter stations may be estimated as a lump sum fixed price for a complete site. For example, a 10 in. meter station with meter, valves, and piping instrumentation may be priced at \$250,000 per site including material and

labor cost. If there are two such meter stations on the pipeline, we would estimate total meter costs at \$500,000.

12.2.6 SCADA and Telecommunication System

This category covers costs associated with Supervisory Control and Data Acquisition (SCADA), telephone, microwave, etc. SCADA system costs include the facilities for remote monitoring, operation, and control of the pipeline from a central control center. Depending upon the length of the pipeline, number of pump stations, valve stations, etc., the cost of these facilities may range from \$2 million to \$5 million or more. An estimate based on the total project cost may range from 2% to 5%.

12.2.7 Engineering and Construction Management

Engineering and construction management consists of preliminary and detailed engineering design costs and personnel costs associated with management and inspection of the construction effort for pipelines, pump stations, and other facilities. This category usually ranges from 15% to 20% of total pipeline project costs.

12.2.8 Environmental and Permitting

In the past, environmental and permitting costs used to be a small percentage of the total pipeline system costs. In recent times, due to stricter environmental and regulatory requirements, this category now includes items such as an environmental impacts report, environmental studies pertaining to the flora and fauna, fish and game, endangered species, sensitive areas such as Native American burial sites, and allowance for habitat mitigation. The latter cost includes the acquisition of new acreage to compensate for areas disturbed by the pipeline route. This new acreage will then be allocated for parks, wildlife preserves, etc.

Permitting costs would include pipeline construction permits such as road crossings, railroad crossings, river and stream crossings, and permitting for antipollution devices for pump stations and tank farms. Environmental and permitting costs may be as high as 10% to 15% of the total project costs.

12.2.9 Right-of-Way Acquisitions

Right of way (ROW) must be acquired for building a pipeline along private lands, farms, public roads, and railroads. In addition to initial acquisition costs there may be annual lease costs that the pipeline company will have to pay railroads, agencies, and private parties for pipeline easement and

maintenance. The annual ROW costs would be considered an expense and would be included in the operating costs of the pipeline. For example, the ROW acquisitions costs for a pipeline project may be \$20 million, which would be included in the total capital costs of the pipeline project. In addition, annual lease payments for ROW acquired may be a total of \$500,000 a year, which would be included with other operating costs such as pipeline O&M, G&A costs, etc.

Historically, ROW costs have been in the range of 6% to 8% of total project costs for pipelines.

12.2.10 Other Project Costs

Other project costs would include allowance for funds used during construction (AFUDC), legal and regulatory costs, and contingency costs. Contingency costs cover unforeseen circumstances and design changes including pipeline rerouting to bypass sensitive areas, pump stations and facilities modifications not originally anticipated at the start of the project. AFUDC and contingency costs will range between 15% and 20% of the total project cost.

12.3 Operating Costs

The annual operating cost of a pipeline consists mainly of the following:
Pump station energy cost (electricity or natural gas)

- Pump station equipment maintenance costs (equipment overhaul, repairs, etc.)

- Pipeline maintenance cost including line rider, aerial patrol, pipe replacements, relocations, etc.

- SCADA and telecommunication costs

- Valve and meter station maintenance

- Tank farm operation and maintenance

- Utility costs: water, natural gas, etc.

- Ongoing environmental and permitting costs

- Right-of-way lease costs

- Rentals and lease costs

- General and administrative costs including payroll

In the above list, pump station costs include electrical energy and equipment maintenance costs, which can be substantial. Consider two pump stations of 5000 HP, each operating 24 hr a day, 350 days a year with a 2 week shutdown for maintenance. This can result in annual O&M costs of \$6 million to \$7 million based on electricity costs of 8 to 10 cents/kWh.

In addition to the power cost other components of O&M costs include annual maintenance and overhead, which can range from \$0.50 million to \$1.0 million depending on the equipment involved.

12.4 Feasibility Studies and Economic Pipe Size

In many instances we have to investigate the technical and economic feasibility of building a new pipeline system to provide transportation services for liquids from a storage facility to a refinery or from a refinery to a tank farm. Other types of studies may include technical and economic feasibility studies for expanding the capacity of an existing pipeline system to handle additional throughput volumes due to increased market demand or refinery expansion.

Grass roots pipeline projects, where a brand new pipeline system needs to be designed from scratch, involve analysis of the best pipeline route, optimum pipe size, and pumping equipment required to transport a given volume of liquid. In this section we will learn how an economic pipe size is determined for a pipeline system, based on an analysis of capital and operating costs.

Consider a project in which a 100 mile pipeline is to be built to transport 8000 bbl/hr of refined products from a refinery to a storage facility. The question is: What pipe diameter and pump stations are optimal for handling this volume?

Let us assume that we selected a 16 in. diameter pipe to handle the designated volume and calculated that this system needs two pump stations of 2500 HP each. The total cost of this system of pipe and pump stations can be calculated and we will call this the 16 in. cost option. If we chose a 20 in. diameter pipeline, the design would require one 2000 HP pump station. In the first case, more HP and less pipe will be required. The 16 in. pipeline system would require approximately 20% less pipe than the 20 in. option. However, the 16 in. option requires 2.5 times the HP required in the 20 in. case. Therefore, the annual pump station operating costs for the 16 in. system would be higher than the 20 in. case, since the electric utility cost for the 5000 HP pump stations will be higher than that for the 2000 HP station required for the 20 in. system. Therefore, to determine the optimum pipe size required, we must analyze the capital costs and the annual operating costs to determine the scenario that gives us the least total cost, taking into account a reasonable project life. We would perform these calculations considering the time value of money, and select the option that results in the lowest present value (PV) of investment.

Generally, in any situation we must evaluate at least three or four different pipe diameters and calculate the total capital costs and operating

costs for each pipe size. As indicated in the previous paragraph, the optimum pipe size and pump station configuration will be the alternative that minimizes total investment, after taking into account the time value of money over the life of the project. We will illustrate this using an example.

Example Problem 12.2

A city is proposing to build a 24 mile long pipeline to transport water at a flow rate of 14.4 million gal/day. There is static elevation head of 250 ft from the originating pump station to the delivery terminus. A minimum delivery pressure of 50 psi is required at the pipeline terminus.

The pipeline operating pressure must be limited to 1000 psi using steel pipe with a yield strength of 52,000 psi. Determine the optimum pipe diameter and the HP required for pumping this volume on a continuous basis, assuming 350 days operation, 24 hr a day. Electricity costs for driving the pumps will be based on 8 cents/kWh. The interest rate on borrowed money is 8% per year. Use the Hazen-Williams equation for pressure drop with a C-factor of 100.

Assume \$700/ton for pipe material cost and \$20,000/inch-diameter-mile for pipeline construction cost. For pump stations, assume a total installed cost of \$1500 per HP. To account for items other than pipe and pump stations in the total cost use a 25% factor.

Solution

First we have to bracket the pipe diameter range. If we consider 20 in. diameter pipe, 0.250 in. wall thickness, the average water velocity using Equation (3.11) will be

$$V = 0.4085 \times 10,000 / 19.5^2 = 10.7 \text{ ft/s}$$

where 10,000 gal/min is the flow rate based on 14.4 million gal/day. This is not a very high velocity. Therefore, 20 in. pipe can be considered as one of the options. We will compare this with two other pipe sizes: 22 in. and 24 in. nominal diameter.

Initially, we will assume 0.500 in. pipe wall thickness for the 22 in. and 24 in. pipes. Later we will calculate the actual required wall thickness for the given MAOP. Using ratios, the velocity in the 22 in. pipe will be approximately

$$10.7 \times (19.5/21)^2 \text{ or } 9.2 \text{ ft/s}$$

and the velocity in the 24 in. pipe will be

$$10.7 \times (19.5/23)^2 \text{ or } 7.7 \text{ ft/s}$$

Thus, the selected pipe sizes of 20 in., 22 in., and 24 in. will result in a water velocity between 7.7 ft/s and 10.7 ft/s, which is an acceptable range of velocities in a pipe.

Next we need to choose a suitable wall thickness for each pipe size to limit the operating pressure to 1000 psi. Using the internal design pressure Equation (4.3), we calculate the pipe wall thickness required as follows:

For 20 in. pipe, the wall thickness is

$$T=1000 \times 20 / (2 \times 52,000 \times 0.72) = 0.267 \text{ in.}$$

Similarly for the other two pipe sizes we calculate:

For 22 inch pipe, the wall thickness is

$$T=1000 \times 22 / (2 \times 52,000 \times 0.72) = 0.294 \text{ in.}$$

For 24 in. pipe, the wall thickness is

$$T=1000 \times 24 / (2 \times 52,000 \times 0.72) = 0.321 \text{ in.}$$

Using the closest commercially available pipe wall thicknesses, we choose the following three sizes:

20 in., 0.281 in. wall thickness (MAOP=1052 psi)

22 in., 0.312 in. wall thickness (MAOP=1061 psi)

24 in., 0.344 in. wall thickness (MAOP=1072 psi)

The revised MAOP values for each pipe size, with the slightly higher than required minimum wall thickness, were calculated as shown within parentheses above. With the revised pipe wall thickness, the velocity calculated earlier will be corrected to

$$V_{20}=10.81 \text{ ft/s}$$

$$V_{22}=8.94 \text{ ft/s}$$

$$V_{24}=7.52 \text{ ft/s}$$

Next, we calculate the pressure drop due to friction in each pipe size at the given flow rate of 10,000 gal/min, using the Hazen-Williams equation with a C-factor of 100. From Equation (3.36) for the 20 in. pipeline

$$10,000 \times 60 \times 24 / 42 = 0.1482 \times 100 (20 - 2 \times 0.281)^{2.63} (P_m / 1.0)^{0.54}$$

Rearranging and solving for the pressure drop P_m we get

$$P_m = 63.94 \text{ psi/mile for the 20 in. pipe}$$

Similarly, we get the following for the pressure drop in the 22 in. and 24 in. pipelines:

$P_m=40.25$ psi/mile for the 22 in. pipe

$P_m=26.39$ psi/mile for the 24 in. pipe

We can now calculate the total pressure required for each pipe size, taking into account the friction drop in the 24 mile pipeline and the elevation head of 250 ft along with a minimum delivery pressure of 50 psi at the pipeline terminus.

Total pressure required at the origin pump station is:

$(63.94 \times 24) + 250 \times 1.0/2.31 + 50 = 1692.79$ psi for the 20 in. pipe

$(40.25 \times 24) + 250 \times 1.0/2.31 + 50 = 1124.23$ psi for the 22 in. pipe

$(26.39 \times 24) + 250 \times 1.0/2.31 + 50 = 791.59$ psi for the 24 in. pipe

Since the MAOP of the pipeline is limited to 1000 psi, it is clear that we would need two pump stations for the 20 in. and 22 in. pipeline cases while one pump station would suffice for the 24 in. pipeline case.

The total BHP required for each case will be calculated from the above total pressure and the flow rate of 10,000 gal/min using Equation (5.18) assuming a pump efficiency of 80%. We will also assume that the pumps require a minimum suction pressure of 50 psi.

$BHP = 10,000 \times (1693 - 50) / (0.8 \times 1714) = 11,983$ for 20 in.

$BHP = 10,000 \times (1124 - 50) / (0.8 \times 1714) = 7833$ for 22 in.

$BHP = 10,000 \times (792 - 50) / (0.8 \times 1714) = 5412$ for 24 in.

Increasing the BHP values above by 10% for installed HP and choosing the nearest motor size, we will use 14,000 HP for the 20 in. pipeline system, 9000 HP for the 22 in. system, and 6000 HP for the 24 in. pipeline system. If we had factored in a 95% efficiency for the electric motor and picked the next nearest size motor we would have arrived at the same HP motors as above.

To calculate the capital cost of facilities, we will use \$700 per ton for steel pipe, delivered to the construction site. The labor cost for installing the pipe will be based on \$20,000 per inch-diameter-mile.

The installed cost for pump stations will be assumed to be \$1500/HP. To account for other cost items discussed earlier in this chapter, we will add 25% to the subtotal of pipeline and pump station cost.

The estimated capital costs for the three pipe sizes are summarized in Table 12.1. Based on total capital costs alone, it can be seen that the 24 in. system is the best. However, we will have to look at the operating costs as well, before making a decision on the optimum pipe size.

Next, we calculate the operating cost for each scenario, using electrical energy costs for pumping. As discussed in an earlier section of this chapter,

Table 12.1 Capital Costs for Varying Pipe Sizes

Capital cost (million \$)	Pipe size		
	20 in.	22 in.	24 in.
Pipeline	2.62	3.21	3.85
Pump stations	21.00	13.50	9.00
Other (25%)	5.91	4.18	3.21
Total	29.53	20.88	16.07

many other items enter into the calculation of annual operating costs, such as O&M, G&A costs, etc. For simplicity, we will increase the electrical cost of the pump stations by a factor to account for all other operating costs.

Using the BHP calculated at each pump station for the three cases and 8 cents/kWh for electricity cost, we find that the annual operating cost for 24 hr operation per day, 350 days per year the annual costs are:

$$11,983 \times 0.746 \times 24 \times 350 \times 0.08 = \$6.0 \text{ million/yr for 20 in.}$$

$$7833 \times 0.746 \times 24 \times 350 \times 0.08 = \$3.93 \text{ million/yr for 22 in.}$$

$$5412 \times 0.746 \times 24 \times 350 \times 0.08 = \$2.71 \text{ million/yr for 24 in.}$$

Strictly speaking, the above costs will have to be increased to account for the demand charge for starting and stopping electric motors. The utility company may charge based on the kW rating of the motor. This will range from \$4 to \$6 per kW/month. Using an average demand charge of \$5/kW/month, we get the following demand charges for the pump station in a 12 month period:

$$14,000 \times 5 \times 12 = \$840,000 \text{ for 20 in.}$$

$$9000 \times 5 \times 12 = \$540,000 \text{ for 22 in.}$$

$$6000 \times 5 \times 12 = \$360,000 \text{ for 24 in.}$$

Adding the demand charges to the previously calculated electric power cost, we get the total annual electricity costs as

$$\$6.84 \text{ million/yr for 20 in.}$$

$$\$4.47 \text{ million/yr for 22 in.}$$

$$\$3.07 \text{ million/yr for 24 in.}$$

Increasing above numbers by a 50% factor to account for other operating costs such as O&M, G&A, etc., we get the following for total annual costs for each scenario:

$$\$10.26 \text{ million/yr for 20 in.}$$

$$\$6.7 \text{ million/yr for 22 in.}$$

\$4.6 million/yr for 24 in.

Next, we use a project life of 20 years and interest rate of 8% to perform a discounted cash flow (DCF) analysis, to obtain the present value of these annual operating costs. Then the total capital cost calculated earlier and listed in Table 12.1 will be added to the present values of the annual operating costs. The present value (PV) will then be obtained for each of the three scenarios.

PV of 20 in. system = \$29.53 + present value of \$10.26 million/yr at 8% for 20 years

or

$PV_{20} = 29.53 + 100.74 = \130.27 million

Similarly, for the 22 in. and 24 in. systems, we get

$PV_{22} = 20.88 + 65.78 = \86.66 million

$PV_{24} = 16.07 + 45.16 = \61.23 million

Thus, based on the net present value of investment, we can conclude that the 24 in. pipeline system with one 6000 HP pump station is the preferred choice.

In the preceding calculations we made several assumptions for the sake of simplicity. We considered major cost components, such as pipeline and pump station costs, and added a percentage of the subtotal to account for other costs. Also, in calculating the PV of the annual costs we used constant numbers for each year. A more rigorous approach would require the annual costs be inflated by some percentage every year to account for inflation and cost of living adjustments. The Consumer Price Index (CPI) could be used in this regard. As far as capital costs go, we can get more accurate results if we perform a more detailed analysis of the cost of valves, meters, and tanks instead of using a flat percentage of the pipeline and pump station costs. The objective in this chapter was to introduce the reader to the importance of economic analysis and to outline a simple approach to selecting the economical pipe size.

In addition, the earlier section on cost of services and tariff calculations provided an insight into how transportation companies finance a project and collect revenues for their services.

12.5 Summary

We have reviewed the major cost components of a pipeline system consisting of pipe, pump station, etc., and illustrated methods of estimating the capital costs of these items. The annual costs such as electrical energy,

O&M, etc., were also identified and calculated for a typical pipeline. Using the capital cost and operating cost, the annual cost of service was calculated based on specified project life, interest cost, etc. We thus determined the transportation tariff that could be charged for shipments through the pipelines. Also a methodology for determining the optimal pipe size for a particular application using present value (PV) was explained. Considering three different pipe sizes, we determined the best option based on a comparison of PV of the three different cases.

12.6 Problems

- 12.6.1 Calculate the annual cost of service and transportation tariff to be charged for shipments through a refined products pipeline as follows. The pipeline is 90 km long, 400 mm diameter and 8 mm wall thickness and constructed of steel with a yield strength of 448 MPa. The terrain is essentially flat. It is used to transport gasoline (specific gravity=0.74 and viscosity=0.65 cSt at 60°F) at a flow rate of 2000 m³/hr. The annual operating costs such as O&M, G&A, etc., are estimated to be \$2 million and do not include power costs. Assume a power cost of \$0.10 per kWh for the electric motor-driven pumps. The project will be financed at a debt/equity ratio 70/30. The interest rate on debt is 7% and the rate of return allowed by regulators is 14%. Assume a project life of 25 years and overall tax rate of 35%.
- 12.6.2 Compare pipeline sizes of 12 in., 14 in., and 16 in. for an application that requires shipments of crude oil from a tank farm to a refinery 30 miles away at 5000 bbl/hr. The tank farm is at an elevation of 350 ft while the refinery is at 675 ft elevation. The pipe MAOP is limited to 1400 psi. Consider 5LX-65 pipe and a 72% design factor. The suction pressure at the tank farm may be assumed at 50 psi and the delivery pressure at the refinery is 30 psi. The pipeline will be operated 355 days a year, 24 hr per day. Electricity cost is 6 cents/kWh.
- 12.6.3 In Problem 12.6.2, assuming the steady-state flow rate of 5000 bbl/hr remains constant for the first 10 years, what is the estimated tariff? Determine the reduction in tariff if the flow rate is increased by 20% for the next 10 years.